AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT249
Practical modern statistics

Final Examination
Spring Semester 2025/2026

Date: 6 /June/2024

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Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed: 120 Minutes.**

Instructions:

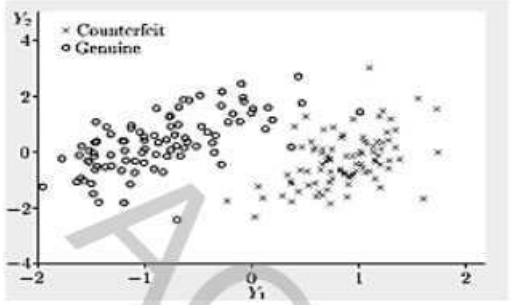
- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

Part1: Q-1 [20 marks]

A- [12 marks] Answer the following 5 questions as True or False (2 marks) and write the correct statement if the answer is false or state why it is false(1mark):

1- In figure1 below, the principal component Y_2, shown on the y axis of the scatter plot, separates the groups very well.

Figure 1



2- For a variable X, the between-groups variance V_b summarizes the spread of the group means of X.

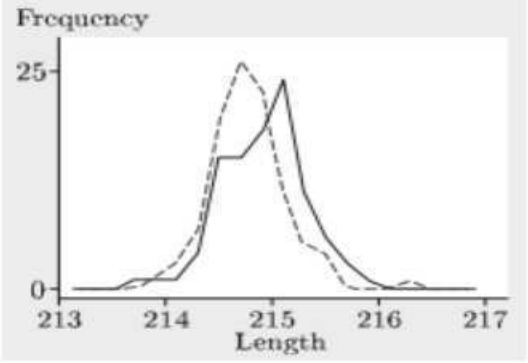
3- The diagonal elements of the correlation matrix are all equal to 1.

4- When multivariate data consist of data from different groups, it is required to calculate between-groups covariances and within-groups covariances.

5- Figure 1 gives the frequency distributions of the length of the counterfeit notes (broken blue line) and the genuine notes (solid black line).

From the graph, it appears to be a clear separation between Lengths of counterfeit notes and genuine notes.

Figure 2



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B- [8 marks] Fill in the blanks in the following 8 statements:

1- ---------- approach is used to find a lower-dimensional approximation to the data when the number of variables is large.
2- ---------- data may be displayed using two or three-dimensional scatterplots.
3- For two events A and B, the conditional probabilities P(A|B) and P(B|A) are related by ----------.
4- A variable that takes ---------- values for different groups is said to separate the groups.
5- The degree to which a linear combination Y captures the total variance is given by the ----------.
6- The correlation matrix of $X_1,...,X_p$ is the covariance matrix of the ---------- variables $Z_1,...,Z_p$).
7- The technique for finding linear combinations D that separates the groups as much as possible is called ----------.
8- A set of points lying close together on a scatterplot is often referred to as a ---------- of points.

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Part2: Each question is worth 20 marks. Answer the following questions:

Q–2: [10+10 marks]:

(a) Table 1 contains a small multivariate data set.

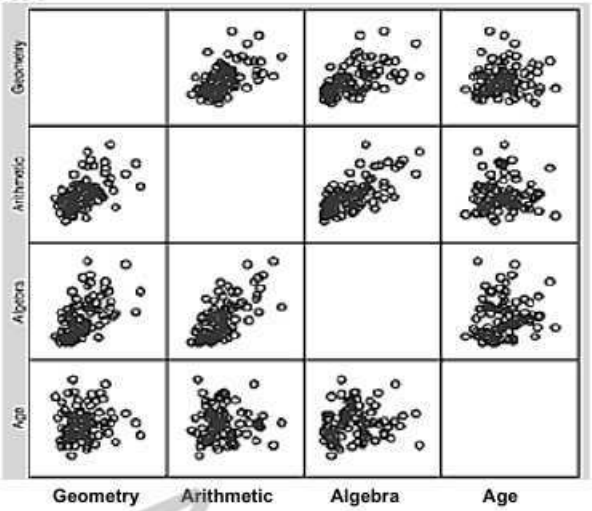
Table 1

X_1	X_2	X_3	X_4	X_5
3064	1201	10	361	2
3000	1053	11	338	3
3155	1133	19	393	2
3085	963	4	467	3
3245	1258	36	294	1
3045	1208	14	371	3

1- State the number of variables and the number of observations in this data set. What are the values of x_{12} and x_{44}?
2- Calculate the covariance between X_1 and X_2. (You may assume that $\bar{x}_1$= 3099 and $\bar{x}_2$ = 1136.)

(b) In the early 1900s, a researcher interested in the theory of reasoning investigated the mathematical ability of boys at an English public school. The boys sat three mathematics exams set by the researcher. One exam focused on geometry, one on arithmetic and the other on algebra. When marking the exams, the researcher focused on 'the intellectual processes involved in answering the questions. Exam scores for 83 boys, aged between 13 and 18, were obtained. A matrix scatterplot of the data is shown in Figure 3.

Figure 3



1- Describe the relationships between the three mathematics variables.
2- For which two variables is the relationship strongest? Explain your answer.
3- How is age related to the other three variables?
4- Comment on why there are no scatterplots in the diagonal positions.

Q–3: [10+10marks]:

(a) Four variables, Y_1, Y_2, Y_3, Y_4, were recorded for a set of 50 observations. The covariance matrix, calculated using these data, is

$$\mathbf{S} = \begin{pmatrix} 224 & 1803 & -2 & 269 \\ 1803 & 30204 & 971 & 3441 \\ -2 & 971 & 229 & -10 \\ 269 & 3441 & -10 & 1774 \end{pmatrix}$$

Use this covariance matrix to answer the following questions.

1- State why it cannot be the correlation matrix for a set of data (state one reason).
2- Which variable had the largest variance?
3- Between which pair of variables was there the greatest covariance (in absolute value)?
4- Does the covariance matrix support the below two assertions? Explain your answers?
First assertion: There is a positive relationship between the variables Y_1 and Y_2.
Second assertion: There does not seem to be a relationship between Y_3 and Y_4.

(b) In a study of heart disease, measurements were taken on 120 patients with heart disease and 150 patients without heart disease. Four of the variables recorded for each patient were age, resting blood pressure, serum cholesterol and maximum heart rate. The mean and standard deviation for the age variable for each group of patients are given in Table 2.

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Table 2

Variable	Heart disease present Mean	SD	Heart disease absent Mean	SD
Age	56.6	8.12	52.7	9.51

1- For the variables age, the grand mean is 54.4333, obtain the between-groups variance and the within-groups variance, and hence calculate the separation between patients with heart disease and patients without heart disease.
2- The separations for the other three variables are given in Table3 below

Table 3

Variable	Separation
Blood pressure	0.024
Cholesterol	0.014
Maximum heart rate	0.211

Which variable gives the best separation between the two groups of patients? Which gives the least separation? Explain your answer.

Q–4: [4+4+4+4+4 marks]:

(a) Descriptions of two multivariate data sets are given below. For each data set, state the dimension p and the number of observations n.

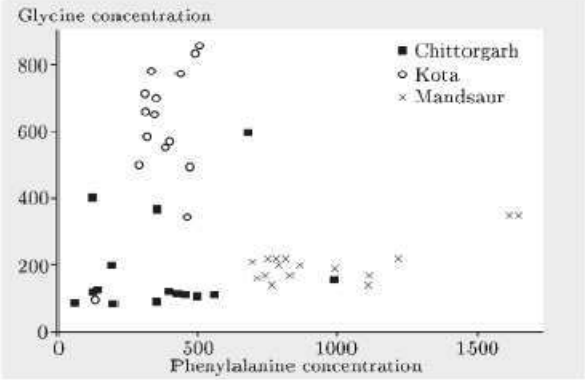
1- In a study on interest rates in different countries, data on twenty countries were collected. For each country, two interest rates were recorded: the average six-month interest rate during 2006, the average three-month interest rate during 2006.
2- A study was conducted to investigate the effect of exercise on the bones of people with osteoporosis. Samples of bone tissue from fifteen patients were studied. The levels of each of five proteins in each sample were measured.

(b) Figure 4 depicts a scatterplot of the concentrations (in g% of total amino acids) of two of the amino acids, glycine and phenylalanine samples of cultivated opium. Fifty of the samples came from three divisions: Chittorgarh (15 samples), Kota (17 samples) and Mandsaur (18 samples).

1- Which division of the cultivated opium samples have higher glycine concentrations? Which of them have higher concentrations of phenylalanine?
2- Which division of the cultivated opium samples can be separated by glycine concentration from others? And which division can be separated by phenylalanine concentration from others?
3- Which samples' locations tend to have similar glycine concentrations?

YOU EXP...

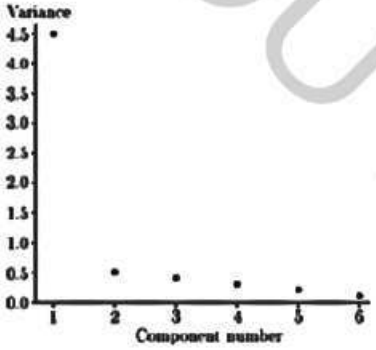
Figure 4



Q–5: [4+4+4+4+4 marks]:

(a) A principal component analysis was carried out on a standardized data set consisting of 96 observations and 6 variables. The resulting scree plot is shown in Figure 5.

Figure 5



1- Based on the scree plot given in Figure 5, how many components should be retained explain your answer.

2- Use Figure 5 to estimate the percentage variance explained by the first component.

(b) The probability for events A_1 and A_2 are $P(A_1) = 0.65$ and $P(A_2) = 0.35$.
It is known that the events A_1 and A_2 are mutually exclusive events.
Suppose $P(A_1 \cap B) = 0.013$ and $P(A_2 \cap B) = 0.0175$.

1- Compute $P(B|A_1)$ and $P(B|A_2)$.

2- Compute $P(B)$.

3- Compute $P(A_1|B)$.

End of questions

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